

Pre-processing

1. Binning / Discretization

Given a set of thresholds $t_1, \dots, t_n$ we map continuous inputs to discrete inputs as

$$X^{\text{BINNED}} \leftarrow 0 \text{ if } x < t_1$$

$$X^{\text{BINNED}} \leftarrow i \text{ if } t_i \leq x \leq t_{i+1}$$

$$X^{\text{BINNED}} \leftarrow n \text{ if } t_n < x$$

2. Normalization / Standardization

$$X^{\text{NORM}} \leftarrow \frac{x - \min_x}{\max_x - \min_x}$$

Force range for all features to be $[0, 1]$.

$$X^{\text{STAND}} \leftarrow \frac{x - \mu_x}{\sigma_x}$$

Force mean zero, standard deviation 1.

... x ... values associated

$\min_x, \max_x, \mu_x, \sigma_x$ are values associated with a feature. Computed on training set.

Example

Medical diagnosis task.

Feature: temp of patient.

Threshold: $t_1 = 37^\circ\text{C}$, $t_2 = 38^\circ\text{C}$

$$X = 40^\circ\text{C} \Rightarrow X^{\text{BIN}} = 2$$

$$X = 36^\circ\text{C} \Rightarrow X^{\text{BIN}} = 0$$

$$X = 37.5^\circ\text{C} \Rightarrow X^{\text{BIN}} = 1$$

Over training set:

$$\min_x = 34^\circ\text{C} \quad \max_x = 42^\circ\text{C}$$

$$\mu_x = 37^\circ\text{C}, \quad \sigma_x = 1^\circ\text{C}$$

$$X = 38^\circ\text{C} \Rightarrow X^{\text{NORM}} = \frac{1}{2}$$

$$X^{\text{STAND}} = 1$$

3. One-hot Encoding

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$$X_i^{\text{ONEHOT}} = \begin{cases} 1 & \text{if } x \in \text{Category } i \\ 0 & \text{if } x \notin \text{Category } i \end{cases}$$

x has length equal to number of categories.

Example

Medical diagnosis task

Blood type: (O, A, B, AB)

$$x = A, \quad x^{\text{ONEHOT}} = [0, 1, 0, 0]$$

$$x = AB, \quad x^{\text{ONEHOT}} = [0, 0, 0, 1]$$

Regularization

Choose parameters to minimize trade-off between L and constraints (magnitude of parameters).

Parameter controlling trade-off

Parameters

$$\tilde{L} = L + \lambda \Omega(\vec{w})$$

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original
loss function

measure of magnitude
of parameters

Typically, Ω is L2 norm

$$\Omega(\vec{w}) = \frac{1}{2} \sum_j w_j^2$$

$$\tilde{L} = L + \frac{1}{2} \lambda \sum_j w_j^2$$

We want to backprop to compute

$$\frac{\partial \tilde{L}}{\partial w_i} = \frac{\partial L}{\partial w_i} + \lambda w_i$$

In L1 regularization

$$\Omega(\vec{w}) = \sum_j |w_j|$$

$$\tilde{L} = L + \lambda \sum_j |w_j|$$

$$\frac{\partial \tilde{L}}{\partial w_i} = \frac{\partial L}{\partial w_i} + \lambda \operatorname{sign}(w_i)$$

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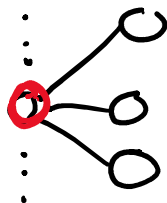
Dropout

An ensemble of neural networks is less likely to overfit.

Train an ensemble of subnetworks of a neural network by randomly masking (multiply output by zero) neurons in network with probability d (dropout rate).

1. All submodels share parameters.
2. Can use different dropout rates for different neurons.

At inference, use full model for prediction. Must multiply weights by $(1-d)$.



At training: $a \cdot w_i \cdot (1 - d)$ average contribution

At inference: $a \cdot w_i \cdot (1 - d)$ average contribution